Nanoparticle trafficking in the tumor immune microenvironment is driven by coupling of cell-intrinsic and extrinsic immune mechanisms

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Motivation

Two Clinical Challenges

THERAPEUTICS

Only ~1% of engineered therapeutic nanoparticles (NP) reach their target in solid tumors, in part due to immune recognition and clearance.

DIAGNOSTICS

Tumors exploit extracellular vesicles (EVs) to modulate the tumor immune microenvironment (TIME), making it difficult to predict patient prognoses

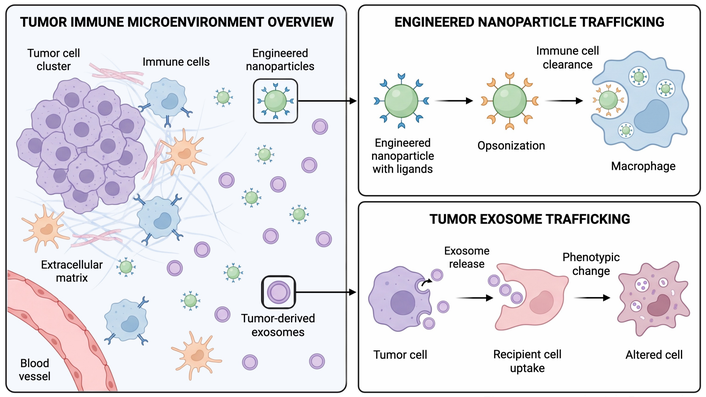


Figure created with Biorender

Hypothesis Nanoparticle trafficking in the TIME is governed by coupling between cell-extrinsic immune-recognition processes and cell-intrinsic biophysical mechanisms of cellular uptake, shaping tissue-level behavior.

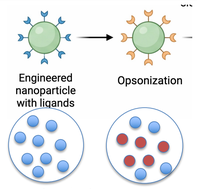
Impact Improved therapeutic design principles and non-invasive diagnostics

Objective Determine how extrinsic factors influence immune recognition and uptake of nanoparticles

Methods

We develop spatially explicit biophysical models of relevant extrinsic and intrinsic phenomena involved in immune recognition and uptake

Complement opsonization



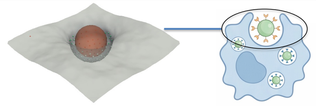
Agent-based simulation model
- Complement tickover, autocatalysis, diffusion, and decay

For more details see [4,5].

Control transport parameters and NP spatial geometry

Measure dynamic complement activity and final opsonized fraction

Phagocytic recognition



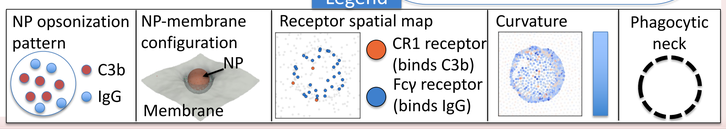
Monte Carlo simulation model
- Deformable membrane surface and asymmetrically opsonized nanoparticle
- Receptor-ligand binding
- Clathrin coat formation

For more model details see [3,6].

Control membrane biophysical properties by changing parameters of the energy functional

Measure receptor engagement, arrangement, and membrane deformation

Legend

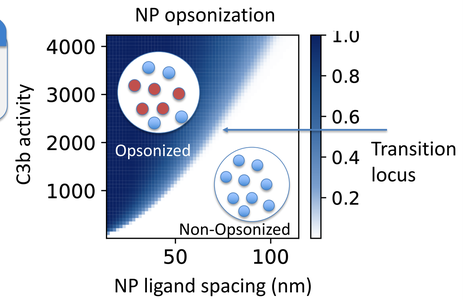


Results

1. Nanoparticle recognition follows a reversible switch-like percolation as a function of extrinsic factors

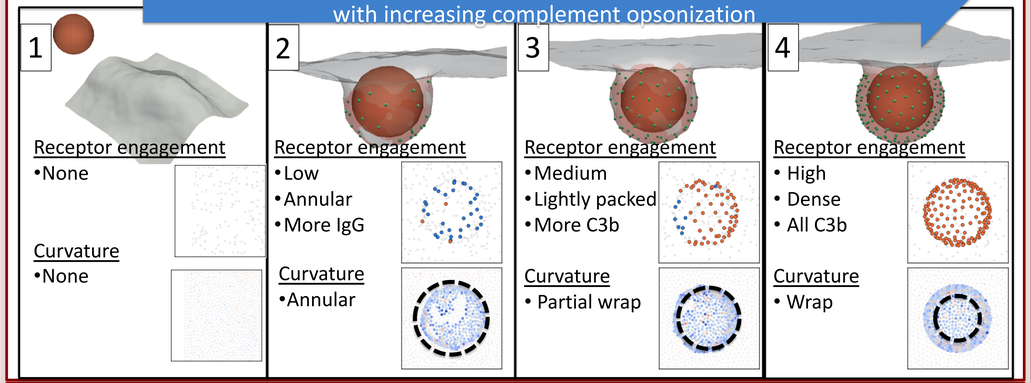
Extrinsic factors influencing opsonization
- C3b activity in TIME: C3b amplification, deactivation, and diffusion
- NP ligand spacing/density

Clinical Implication
1. Small shifts in immune microenvironment may trigger dramatic shifts in complement response to therapies
2. These findings motivate engineering therapies informed by patient-specific biomarkers

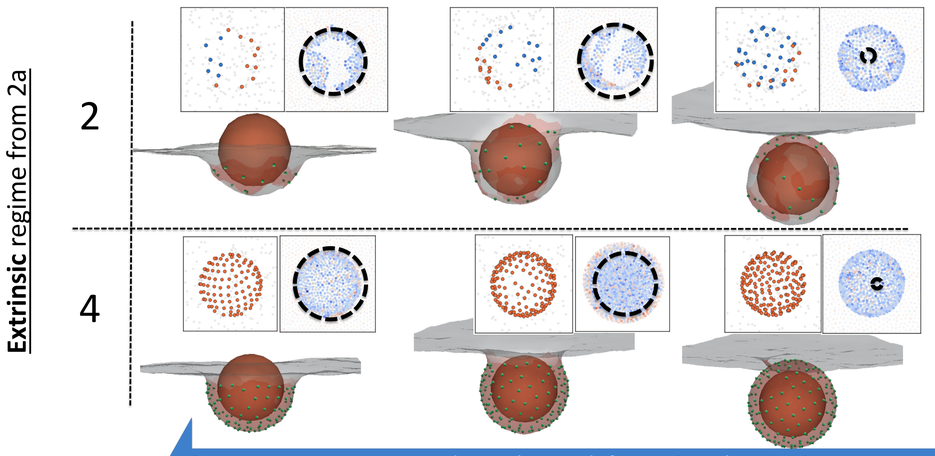


2a. Four distinct regimes of spatial immune receptor engagement based on NP

Spatial receptor engagement and membrane deformation changes with increasing complement opsonization



2b. Immune engagement and deformation further depends on coupling of extrinsic + intrinsic factors



Receptor engagement and membrane deformation changes in response to **intrinsic** cytoskeleton remodeling

Clinical Implication
1. Heterogeneous membrane remodeling in the TIME influences NP clearance and exosome-mediate signal transduction
2. Membrane biomarker profiling could help customize NP treatments and contextualize exosome surface signatures

Conclusions and Impact

CONCLUSIONS

The coupling of cell-extrinsic (complement activity and nanoparticle surface chemistry) and cell-intrinsic (membrane and receptor properties) factors drive immune receptor engagement and membrane remodeling

SCIENTIFIC IMPACT

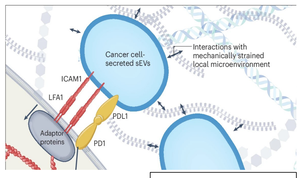
Our work delineates key physical mechanisms governing NP recognition and uptake under heterogeneous immune and biophysical conditions

CLINICAL IMPACT

These findings motivate integrating immune, membrane, and single EV profiling biomarkers to inform patient-specific diagnostics and therapy optimization to tackle inter- and intra- patient heterogeneity

These physics may govern heterogeneous signaling responsible for
1.) malignant and metastatic transformation via exosome trafficking and
2.) reduced therapeutic efficacy in tumors

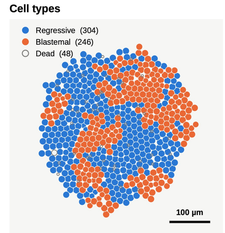
Future Work



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Study tumor immune evasion by modeling EVs with surface PDL1 engaging with T-cell receptors

Visualize interface with STORM microscopy



Study effect of NP and EV uptake on tumor heterogeneity using tissue-level simulation

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